

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. **Project No.:** WW-4705R
Project Name: Hilo WWTP Phase 1 **Submittal Number:**
Submittal Title: For Information Only
TO:
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec:	Paragraph:
Authored By:	Date Submitted:

Submittal Certification	
Check Either (A) or (B): ☐ (A) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> . ☐ (B) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.	
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: <i>Nitesh N</i>	
Printed Name and Title:	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM: Date Received PM/CM to Reviewer: Date Received Reviewer to PM/CM: Date Sent PM/CM to GC:

Nan, Inc

 PROJECT: HILO WWTP REHABILITATION
 AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

 THIS SUBMITTAL HAS BEEN CHECKED BY
 THIS CONTRACTOR. IT IS CERTIFIED
 CORRECT, COMPLETE, AND IN
 COMPLIANCE WITH CONTRACT
 DRAWINGS AND SPECIFICATIONS. ALL
 AFFECTED CONTRACTORS AND
 SUPPLIERS ARE AWARE OF, AND WILL
 INTEGRATE THIS SUBMITTAL (UPON
 APPROVAL) INTO THEIR OWN WORK.

 DATE RECEIVED _____
 SPECIFICATION SECTION # _____
 SPECIFICATION _____
 PARAGRAPH _____
 DRAWING _____
 SUBCONTRACTOR _____
 SUPPLIER _____
 MANUFACTURER _____

 CERTIFIED BY CQCM or Designee : *[Signature]*

Review Comments:

1. Fabricast Valve is not a listed manufacturer for knife gate valves. Contractor shall provide a listed manufacturer per Section 15115-2.04.A.

Response - The valve submitted is "or equal" per Section 15115-2.04.A.



C&B PIPING (W), INC.

P.O. BOX 974

LEEDS, AL 35094

PHONE: (205) 699-0455

FAX: (205) 699-0773

www.cbpipe.com

DATE: 8/22/25

CONTRACTOR: NAN, INC.

PROJECT: HILO WWTP

BID DATE: 8/15/2024

KNIFE GATE VALVE SUBMITTAL

SPEC SECTION 15115-4

AIS DOMESTIC MATERIAL

C&B PIPING WILL PROVIDE LAYOUT DRAWING SUBMITTALS SHOWING PIPE SYSTEMS AND ACTUAL PIPE SIZES, CLASSES, JOINTS PROPOSED FOR SUBMITTAL APPROVAL. THE DRAWINGS WILL BE SUBMITTED BY SYSTEM AS THEY ARE COMPLETED. THE SUBMITTAL PRODUCT LISTING AND ATTACHED CUT SHEETS ARE FOR REFERENCE TO COVER THE ITEMS THAT WILL BE SHOWN ON THE PIPING LAYOUT DRAWINGS.

KNIFE GATE VALVE

316SS AIS DOMESTIC

1) WAFER FLANGED STYLE 316SS KNIFE GATE VALVE

FABRICAST

NOTE: SUPPLIED WITH GEAR'D HAND WHEEL ACTUATOR

NOTE: 24" KNIFE GATE VALVES WILL BE SUPPLIED W/TORQUE TUBE VSE

2.04 KNIFE GATE VALVES (GV60)

A. Manufacturers: One of the following or equal:

1. Dezurik Hilton, H-200 Series.
2. ITT Fabri-Valve, C37.

B. Design:

1. Valves shall conform to the requirements for design, leakage and production testing of AWWA C520.
2. Size and configuration: Indicated on the Drawings.
3. Suitable for service under pressures equal to and less than 150 pounds per square inch gauge.
4. Full round port, resilient seated, raised face design.
5. Flanged wafer design drilled and tapped to ASME Class 125/150 standard. Face-to-face dimension shall be between 4 to 4-1/2 inches.
6. Rounded gate with beveled edge, finish-ground to 32 RMS, maximum, on both sides. The bottom edge shall have a beveled knife edge.
7. Body to incorporate guides and jams to assist in seating.
8. Packing shall consist of multiple layers of packing.
9. Materials:
 - a. Body: cast Type 316 stainless steel or 2205 Duplex.
 - b. Bonnet (if applicable): Type 316 stainless steel or 2205 Duplex.
 - c. Wetted components (including gate): Type 316 stainless steel or 2205 Duplex.
 - d. Seat: NBR, EPDM, or Viton™.
 - e. Yoke sleeve: Acid resisting bronze or aluminum bronze.
 - f. Packing: PTFE.
 - g. Packing gland: Match valve body material.
 - h. Yoke: Type 316 stainless steel.
 - i. Fasteners: Shall be stainless steel.
10. Outside screw and yoke (OS&Y) with handwheel operator.
11. Extended stem is required for two 24" knife gate valves as indicated on the Drawings.

2.05 ~~GLOBE AND ANGLE VALVES~~

~~B. General purpose globe and angle valves:~~

~~1. Valves 3 inches and smaller:~~

~~a. Manufacturers: One of the following:~~

~~1) Except in welded steel piping:~~

~~a) Crane, No. 1 Globe or No. 2 Angle.~~

~~b) Lunkenheimer Company Figure No. 2140 Globe or No. 2141 Angle.~~

~~2) In Welded steel piping:~~

~~a) Crane, No. 351.~~

~~b) Lunkenheimer Company, Figure No. 1123; or equal with flanged ends.~~

~~b. Design:~~

~~1) Size and configuration: Indicated on the Drawings.~~

~~2) Valve: Class 125 threaded ends, rated for 250 degrees Fahrenheit at pressure of 170 pounds per square inch.~~

2. Valves larger than 3 inches:
 - a. Manufacturers: One of the following:
 - 1) Crane, No. 351.
 - 2) Lunkenheimer Company, Figure No. 1123.
 - b. Design:
 - 1) Size and configuration: Indicated on the Drawings.
3. Plug disc globe valves: For throttling or for severe service:
 - a. Manufacturers: One of the following:
 - 1) Crane, No. 212P or No. 384P.
 - 2) Lunkenheimer Company, 72-PS or 73-PS1.
 - b. Design:
 - 1) Size and configuration: Indicated on the Drawings.
 - 2) Material: Iron body stainless steel trimmed plug type seat and disc.

2.06 HOSE VALVES:

- A. Hose valves:
 1. Manufacturers:
 - a. Globe threaded valve: One of the following or equal:
 - 1) Crane, No. 7TF.
 - 2) Stockham, Figure No. B22T.
 - b. Angle threaded valve: One of the following or equal:
 - 1) Crane, No. 17TF.
 - 2) Stockham, Figure No. B222T.
 2. Design:
 - a. Size and configuration: Indicated on the Drawings.
 - b. Valve: Globe or angle valve with threaded ends.
 - c. Disc: Renewable, made of Teflon™ or Buna-N.
 - d. Threaded ends rated for a pressure of 200 pounds per square inch.

PART 3 EXECUTION

3.01 COMMISSIONING

- A. As specified in Section 01756 - Commissioning and this Section.
- B. Manufacturer services: For valves larger than 16-inches:
 1. Provide certificates:
 - a. Manufacturer's Certificate of Installation and Functionality Compliance.
- C. Functional testing:
 1. Valves:
 - a. Conduct pressure and leak test as specified in Section 15110 - Common Work Results for Valves.

END OF SECTION



Section 00 48 14



American Iron and Steel Certification

Company Name: Fabricast Valve
Company Address: 1061 Industrial Way
City, State, Zip: Longview, WA 98632

Date: 8/15/2025

Subject: AIS Step Certification for C&B Piping (W) Inc.

I, Sid Somers, certify that the melting, bending, coating, galvanizing, cutting, etc. process for manufacturing or fabricating the following products and/or materials shipped or provided for the subject project is in full compliance with the American Iron and Steel requirement as mandated in EPA's State Revolving Fund Programs.

Valve/product being provided:

24" Figure F40-A6E-DN2 Knife Gate Valve
--Fabricated body with 316 SS flanges and gussets
--MSS Face to Face
--150 PSI CWP design
--Flanges drilled and tapped to ANSI B16.5/150
--316 SS gate
--316 SS metal seat w/ EPDM o-ring seal for uni-directional tight shut off
--TLSP packing rated to 500° F, 3-11 PH
--316 SS packing gland with 304 SS bolts
--Fabricated 316 SS yoke
--5' 316 SS stem extension per detail 'E' of the ext composite dwg
--316 SS non rising stem adapter
--316 SS 2" square drive nut

Serial # 2502941-1 & 1-2

Valve fabrication will takeplace at the following location: **1061 Industrial Way, Lonview, WA 98632**

If any of the above mentioned compliance statements change while providing material to this project we will immediately notify the prime contractor and the engineer.

All listed valve parts / components manufactured by Fabricast valve at above location

Signed: Sid Somers



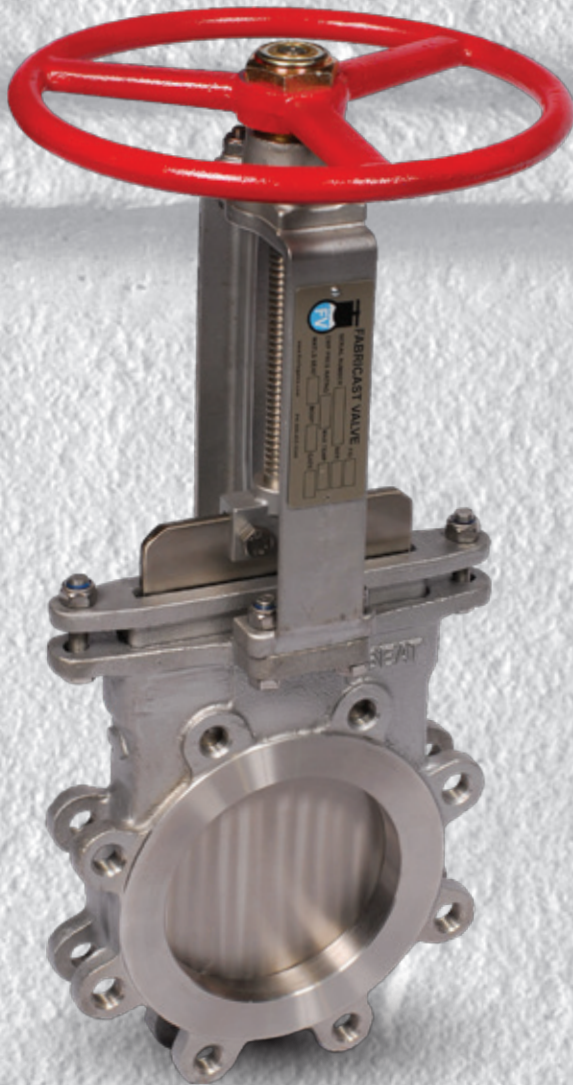
Fabricast Valve Company
1061 Industrial Way, Longview, WA 98632
Toll-Free 1-877-264-2503
Local (360) 425-0306 **Fax** (360) 425-0353
www.knifegates.com
sales@fabricastvalve.com



FIG 40

FIG 40

Fabricast Valve



The Fabricast FIG 40 is a metal seated, uni-directional knife gate valve. Designed and tested to MSS SP81 and ASME B16.5 specifications.

Features

- Heavy Duty Cast CF8 304, CF8M T316 or CG8M T317 body and packing gland. Cast CF8 T304 yoke, stemnut retainer and gate clevis.
- Full Port.
- 150 Psi Cwp Rated.
- Precision Cnc Machined Valve Body And Stem.
- ASTM A 240 304, 316, and 317 Stainless Steel gate fully machined, beveled and surface ground. All edges are chamfered to extend packing life and to provide a superior atmospheric seal.
- 14" and larger valves are supplied with a ball bearing yoke bushing to reduce handwheel rimpull.
- Braided Teflon Synthetic Packing is standard 3-11 PH rating, 500°F.
- Cast WCB Steel handwheel (weldable)
- Easily adapted to Air Cylinder, Hydraulic Cylinder, Bevel Gear, Lever or Electric Motor Operator (no welding required).

Options

- O-ring Seat - Buna N 200 Deg F, Epdm 300 Deg F, Hdpe/uhmwpe 200 Deg F, Fkm/viton - 400 Deg F. Other - Contact Factory
- Other Alloys - 310, 316, 317L, Duplex 2205, 2507, 904L, Hastelloy, Titanium Other - Contact Factory
- Mechanical Lockouts - Manual And Actuated Valves
- Osha Gate Guard
- Pneumatic, Hydraulic And Spring Cylinder, Electric Motor Operator, 3:1 And 4:1 Bevel Gear Operator, Chainwheel, Stem Extensions, Lever Operator
- Packings - Ptfе, Graphite, Kevlar, Food Grade, Other - Contact Factory
- Purge / Flushing Ports
- Limit / Proximity Switches, Solenoid Valves, Positioners, Air Sets, Speed Controls, Exhaust Mufflers, Air Tank Failsafe Systems, Air Boosters
- V Port Inserts For Flow Control
- Deflector Cones For Abrasive Service
- Ni-resist Alloy Stemnut
- Live Loaded Packing System
- Specialty Gate Coatings
- Hardsurfacing For Abrasive Service
- Backing Ring For Reverse Flow Or Horizontal Valve Orientation
- High Pressure
- High Temperature

Previous Model FV-150-MM

Optional Operators see www.knifegates.com for details



Cylinder



Bevel Gear



Electric Motor



Chain Wheel

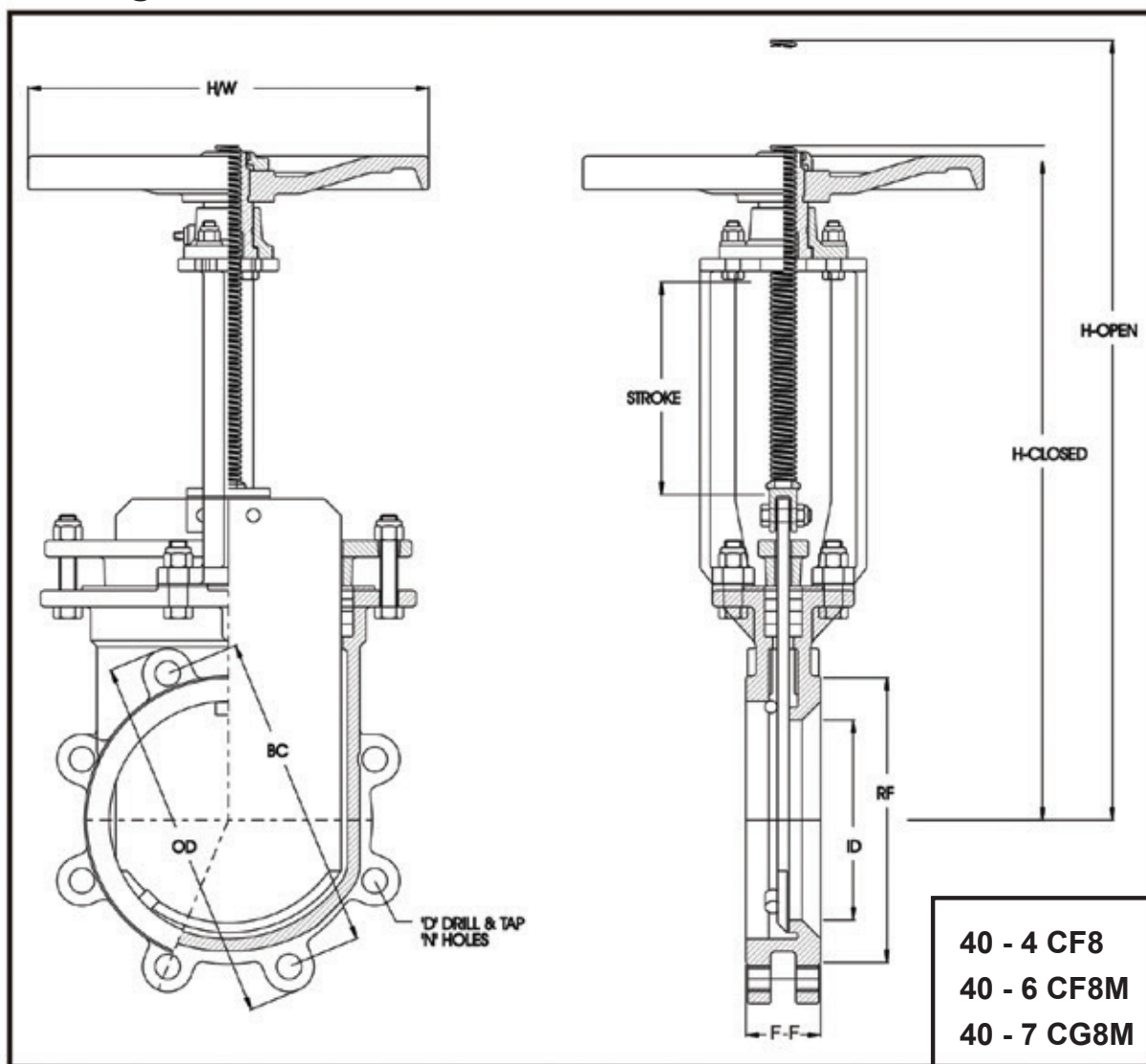


Lever



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 sales@fabricastvalve.com

FIG 40



40 - 4 CF8
40 - 6 CF8M
40 - 7 CG8M

Size		2"	3"	4"	6"	8"	10"	12"	14"	16"	18"	20"	24"	30"	36"
Face to Face	F-F	1 ⁷ / ₈ "	2"	2"	2- ¹ / ₄ "	2- ³ / ₄ "	2- ³ / ₄ "	3"	3"	3- ¹ / ₂ "	3- ¹ / ₂ "	4- ¹ / ₂ "	4- ¹ / ₂ "	5 ⁵ / ₈ "	6"
Flange OD	OD	6"	7- ¹ / ₂ "	9"	11"	13- ¹ / ₂ "	16"	19"	21"	23- ¹ / ₂ "	25"	27- ¹ / ₂ "	32"	38- ³ / ₄ "	46"
Bolt Circle	BC	4- ³ / ₄ "	6"	7- ¹ / ₂ "	9- ¹ / ₂ "	11- ³ / ₄ "	14- ¹ / ₄ "	17"	18- ³ / ₄ "	21- ¹ / ₄ "	22- ³ / ₄ "	25"	29- ¹ / ₂ "	36"	42- ³ / ₄ "
Inside Diameter	ID	2"	3"	4"	6"	8"	10"	12"	13- ⁵ / ₈ "	15- ¹ / ₂ "	17- ¹ / ₂ "	19- ¹ / ₂ "	23- ³ / ₈ "	29- ¹ / ₄ "	35 ³ / ₈ "
Number of Holes	N	4	4	8	8	8	12	12	12	16	16	20	20	28	32
Hole D&T Size	D	⁵ / ₈ "-11	⁵ / ₈ "-11	⁵ / ₈ "-11	³ / ₄ "-10	³ / ₄ "-10	⁷ / ₈ "-9	⁷ / ₈ "-9	1"-8	1"-8	1- ¹ / ₈ "-7	1- ¹ / ₈ "-7	1- ¹ / ₄ "-7	1- ¹ / ₄ "-7	1 ¹ / ₂ "-6
C/L to Top Closed	H-Closed	13- ⁵ / ₈ "	16"	17- ³ / ₄ "	21- ⁵ / ₈ "	26- ³ / ₄ "	32- ¹ / ₈ "	36- ³ / ₈ "	42"	46- ³ / ₄ "	52- ³ / ₈ "	49- ³ / ₄ "	55- ¹ / ₈ "	80"	111- ³ / ₈ "
C/L to Top Open	H-OPEN	16"	19- ³ / ₄ "	21- ⁵ / ₈ "	27- ³ / ₄ "	34- ³ / ₄ "	42- ³ / ₄ "	48- ⁵ / ₈ "	55- ¹ / ₄ "	63- ¹ / ₈ "	70- ¹ / ₂ "	69- ³ / ₄ "	79- ⁷ / ₈ "	114- ¹ / ₄ "	145- ³ / ₄ "
Handwheel Dia.	H/W	8"	8"	8"	12"	12"	15"	15"	15"	20"	20"	20"	30"	18"	18"
Raised Face Diameter	RF	3- ⁵ / ₈ "	5"	6- ³ / ₈ "	8- ¹ / ₂ "	10- ⁵ / ₈ "	12- ³ / ₄ "	15"	16- ¹ / ₄ "	18- ¹ / ₂ "	21"	23"	27- ¹ / ₄ "	33 ³ / ₄ "	40 ¹ / ₄ "
Weight (lbs.)		20	29	38	52	85	120	160	212	300	350	480	600	1136	2205

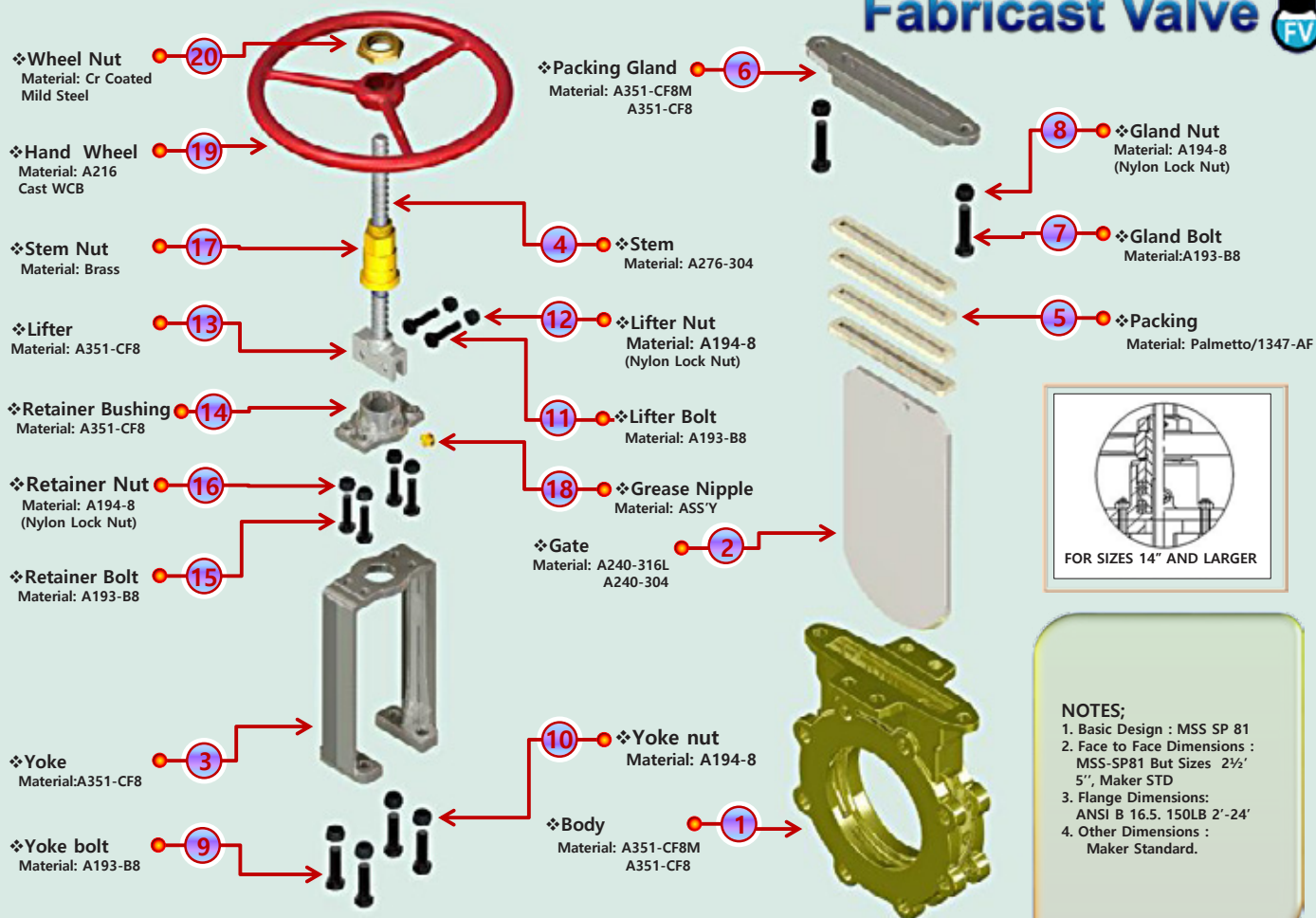
Note: 30" and 36" cast valves with gear operator

Fabricast Valve
Knife Gate Valves & Custom Fabrication / phone (360) 425-0306, fax (360) 425-0353



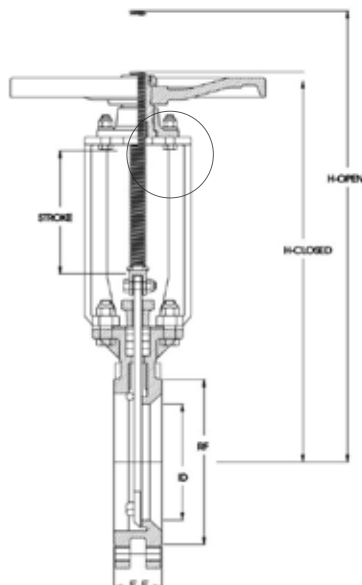
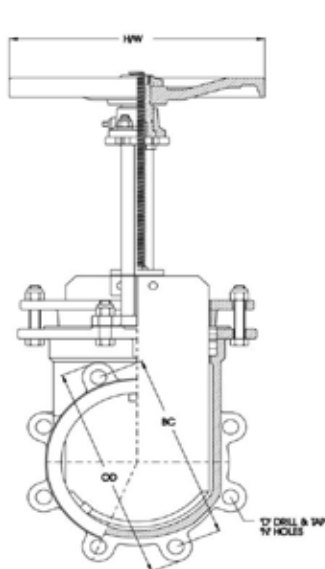
Fabricast Valve Company
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Fabricast Valve 

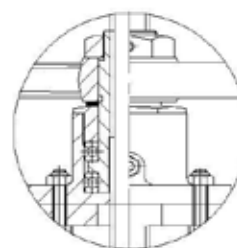




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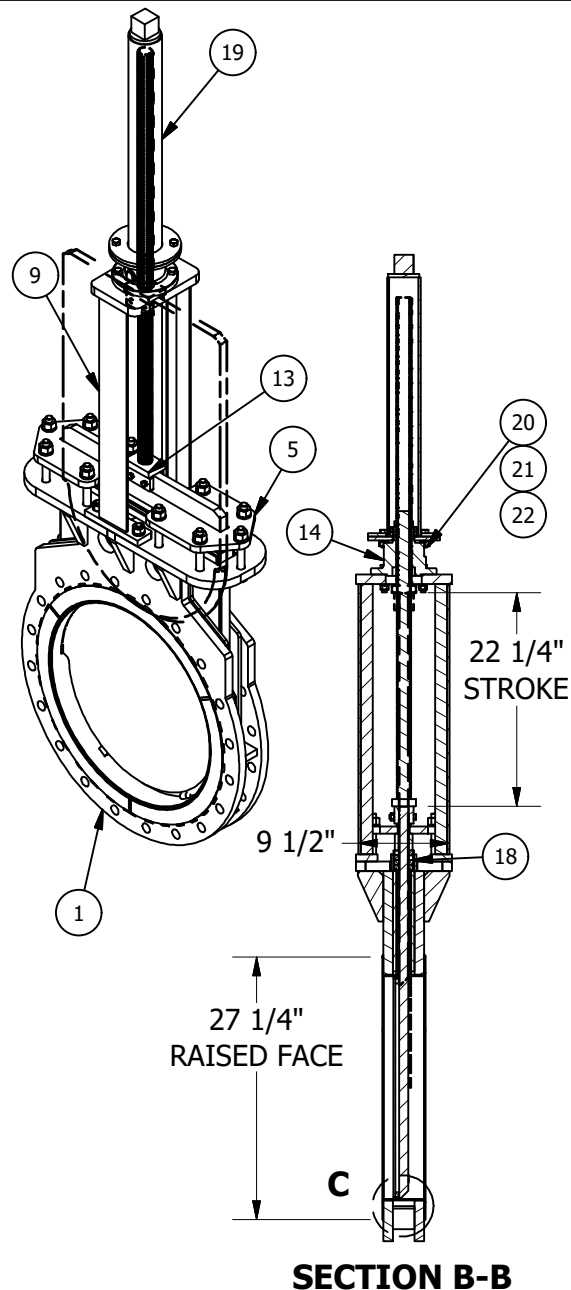
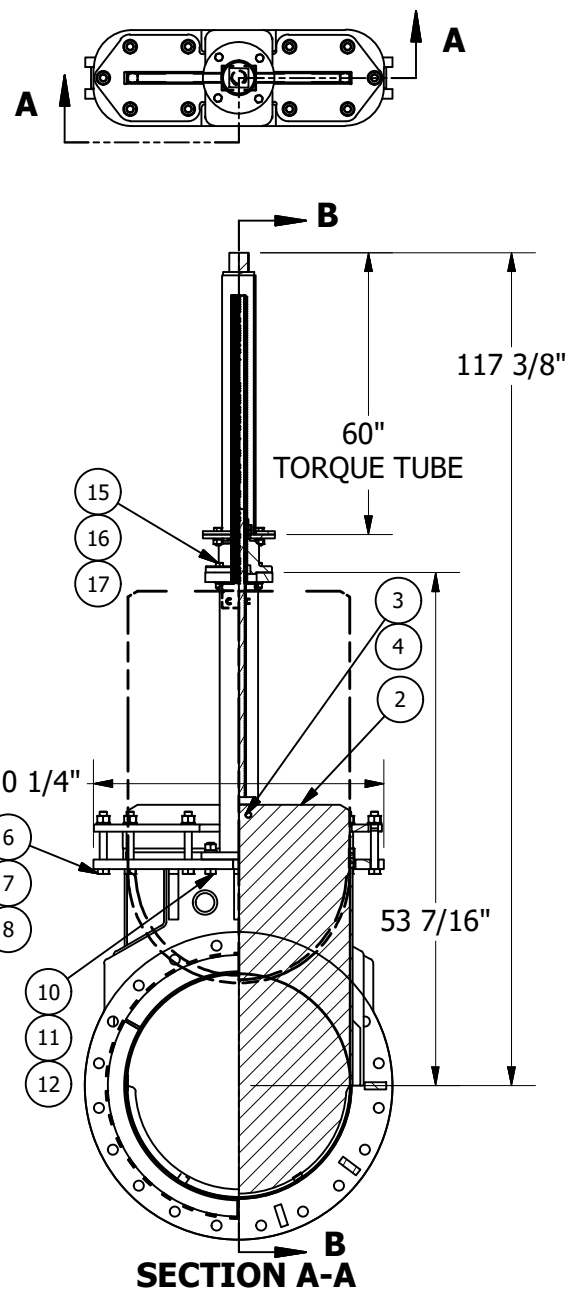


FOR SIZES 14" AND LARGER

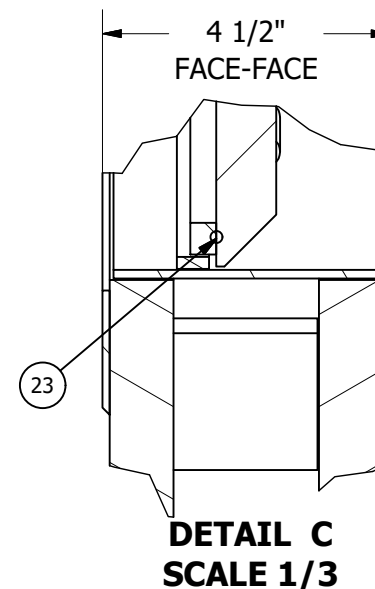
30" & 36" COME
 STANDARD WITH
 GEAR OPERATOR

Dimensions (inches)&Weights (lbs.)

Size	2"	3"	4"	6"	8"	10"	12"	14"	16"	18"	20"	24"	30"	36"
Face to Face F-F	1-7/8"	2"	2"	2-1/4"	2-3/4"	2-3/4"	3"	3"	3-1/2"	3-1/2"	4-1/2"	4-1/2"	5-3/16"	6"
Flange OD	6"	7-1/2"	9"	11"	13-1/2"	16"	19"	21"	23-1/2"	25"	27-1/2"	32"	38-3/4"	46"
Bolt Circle BC	4-3/4"	6"	7-1/2"	11-3/4"	9-1/2"	14-1/4"	17"	18-3/4"	21-1/4"	22-3/4"	25"	29-1/2"	36"	42-3/4"
Inside Diameter ID	2"	3"	4"	6"	8"	10"	12"	13-5/8"	15-1/2"	17-1/2"	19-1/2"	23-3/8"	29-1/4"	35-3/16"
Number of Holes N	4	4	8	8	8	12	12	12	16	16	20	20	28	32
Tap D	5/8"-11	5/8"-11	5/8"-11	3/4"-10	3/4"-10	7/8"-9	1"-8	1"-8	1"-8	1-1/8"-7	1-1/8"-7	1-1/4"-7	1-1/4"-7	1-1/2"-6
C/L to Top Closed H-C	13-5/8"	16"	17-3/4"	21-5/8"	26-3/4"	32-1/8"	36-3/8"	42"	46-3/4"	52-3/8"	76-1/4"	91-1/4"	80"	111-3/8"
C/L to Top Open H-O	16"	19-3/4"	21-5/8"	27-3/4"	34-3/4"	42-3/4"	48-5/8"	55-1/4"	63-1/8"	70-1/2"	96-3/4"	116"	114-1/4"	145-3/4"
Handwheel Dia H/W	8"	8"	8"	12"	12"	15"	15"	15"	20"	20"	20"	30"	18"	18"
Raised Face Dia RF	3-5/8"	5"	6-3/16"	8-1/2"	10-5/8"	12-3/4"	15"	16-1/4"	18-1/2"	21"	23"	27-1/4"	33-3/4"	40-1/4"
Weight (lbs)	20	29	38	52	85	120	160	212	300	350	480	600	1136	2205



PARTS LIST				
ITEM	NAME	DESCRIPTION	PART NUMBER	QTY
1	BODY	FABRICATED - 316 SS	30-073-BODY MACH A	1
2	GATE	1" THK. - 316 SS	14-079-GATE	1
3	GATE BOLT	3/8"-16 UNC X 2-1/2"	HEX BOLT - 316 SS	2
4	GATE NUT	3/8"-16 UNC	LOCK NUT - 316 SS	2
5	GLAND	FABRICATED - 316 SS	40-064-GLAND WELD	1
6	GLAND STUD	3/4"-10 UNC - 6-1/2"	ALL-THREAD - 316 SS	10
7	GLAND WASHER	3/4"	LOCK WASHER - 316 SS	10
8	GLAND NUT	3/4"-10 UNC	HEX NUT - 316 SS	20
9	YOKE	FABRICATED - 316 SS	50-065-YOKE WELD	1
10	YOKE BOLT	3/4"-10 UNC X 2-3/4"	HEX BOLT - 316 SS	4
11	YOKE WASHER	3/4"	LOCK WASHER - 316 SS	4
12	YOKE NUT	3/4"-10 UNC	HEX NUT - 316 SS	4
13	STEM ASSEMBLY	FABRICATED - 316 SS STEM	60-067-LIFTER WELD	1
14	RETAINER	CAST	CAST RETAINER 1.5	1
15	RETAINER BOLT	1/2"-13 UNC X 2-1/2"	HEX BOLT - 316 SS	4
16	RETAINER WASHER	1/2"	LOCK WASHER - 316 SS	4
17	RETAINER NUT	1/2"-13 UNC	HEX NUT - 316 SS	4
18	PACKING	5/8" SQ. X	SQUARE	4
19	N.R.S.A.	FABRICATED - 316 SS	NR0014B	1
20	NRS BOLT	1/2"-13 UNC X 1-1/2"	HEX BOLT - 316 SS	4
21	NRS WASHER	1/2"	LOCK WASHER - 316 SS	4
22	NRS NUT	1/2"-13 UNC	HEX NUT - 316 SS	4
23	O-RING	.275" RND	EPDM	1



C	TORQUE TUBE CHANGED TO 60"	STOSH Z.	8/12/2025
B	DECREASED NRS TOP HEIGHT	STOSH Z.	7/24/2025
A	INITIAL RELEASE	STOSH Z.	5/20/2025
REV	DESCRIPTION	AUTHOR	DATE
REVISION HISTORY			

-NOTE-
UNLESS OTHERWISE SPECIFIED:
1. DIMENSIONS ARE IN INCHES.
2. ALL SMALL FILLETS = R 0.02-0.04.
3. MACH. SURFACE ROUGHNESS = 125.
4. REMOVE ALL BURRS & BREAK ALL SHARP CORNERS/EDGES.

TOLERANCES:

x/x"	+/- 1/16
x.x"	+/- 0.13
x.xx"	+/- 0.06
x.xxx"	+/- 0.031
<	+/- 0.5°

F-24LF401-23.25ID-4.5RF-5FTNRST C

FABRICAST VALVE
1061 Industrial Way, Longview, WA 98632
Phone: 360-425-0306 Fax: 360-425-0353
www.knifegates.com

24" FIGURE F40

DRAWN:	STOSH Z.	05/20/2025
SIZE	A	SCALE 1/20 SHEET 1 OF 1